



Proving Techniques (Part 2)

Today's Agenda

- Why study Proof Techniques
- Proofs we have seen
- Rules of Inference
- Direct Proof
- Indirect Proof
- Fallacies

Motivation

**“The greatest pleasure in life is doing
what people say you cannot do.”**

Hilbert's Noble Dream

Mathematician David Hilbert once proposed a mathematical model that will

- Formulate all of the mathematics
- Complete - Can prove all true statement
- Consistent - Lack of Contradictions
- Decidable - Can prove or disprove any statement



David Hilbert
(1862 - 1943)

Hilbert's Failed Dream

Spoiler Alert: No such system exists. Any mathematical system

- Cannot formulate all true statements. True statement exists outside all of formalism.
- Incomplete - some statements cannot be proven or disproven
- Undecidable - No single algorithm can decide truth value of all statements.

By studying proof techniques, we can know why Hilbert's dream was crushed.

Why Study Proof Techniques

- **Ensures Rigor:** Removes ambiguity with precise justification
- **Develops Reasoning:** Trains logical and critical thinking
- **Validates Truth:** Proves statements beyond examples
- **Builds Formal Frameworks:** Helps categorize, connect, and verify mathematical ideas
- **Avoids Common Mistakes:** e.g. logical fallacies.
- **Enables Program Verification:** Powers computer-assisted proofs and correctness checks

Prerequisites

- n is even $\rightarrow n = 2k$
- n is odd $\rightarrow n = 2k + 1$ or $n = 2k - 1$
- If 2 even number $\rightarrow a = 2k_1$ and $b = 2k_2$ (Never take same variable)
- If 2 odd number $\rightarrow a = 2k_1 + 1$ and $b = 2k_2 + 1$ (Never take same variable)

Can we take two consecutive odd numbers like $2k + 1$ and $2k - 1$?

Prerequisites

- n is divisible by 2 $\rightarrow n = 2k$
- n is divisible by 3 $\rightarrow n = 3k$
- n is divisible by 5 $\rightarrow n = 5k$
- n leaves remainder 1 when divided by 3 $\rightarrow n = 3k + 1, k \in \mathbb{Z}$

Rules of Inference

Rules of Inference - Modus Ponens

Rule:

Premise 1: If p , then q

Premise 2: p

Conclusion : Therefore, q .

If $p \rightarrow q$ and p , then q .

Example:-

Premise 1: *If it is raining, then the ground will be wet and*

Premise 2: *It is raining*

Then: *The ground will be wet*

Rules of Inference - Modus Ponens

Premise 1: *If you study hard, then you will pass the exam.*

Premise 2: *You study hard.*

Conclusion: *Therefore...?*

Rules of Inference - Modus Tollens

Rule:

Premise 1: If p , then q

Premise 2: Not q

Conclusion : Therefore, Not p .

If $p \rightarrow q$ and $\sim q$, then $\sim p$.

Example:

Premise 1: *If it is raining, then the ground will be wet* and

Premise 2: *The ground is **not** wet*

Then: *If it is not raining*

Rules of Inference - Modus Tollens

Premise 1: *If you study hard, then you will pass the exam.*

Premise 2: *You didn't study hard.*

Conclusion: *Therefore...?*

Law of Syllogism (Hypothetical Syllogism)

Rule:

Premise 1: If p , then q

Premise 2: : If q , then r

Conclusion : Therefore, if p then r .

Example:

Premise 1: *If you work out, then you will get stronger and*

Premise 2: *If you get stronger, then you will be able to lift more weight.*

Then: *If you work out, then you will be able to lift more weight.*

Disjunctive Syllogism

Rule:

Premise 1: $p \text{ or } q$

Premise 2: $\neg p$

Conclusion : Therefore q .

Example:

Premise 1: *The book is either on the table or on the shelf and*

Premise 2: *The book is not on the table.*

Then: *The book is on the shelf.*

Problem

Premise 1: *If the alarm goes off, then the house is not secure.*

Premise 2: *The house is secure.*

Conclusion: *Therefore, the alarm did not go off.*

Take a look at this argument and see if you can identify which rule of inference it follows.

Problem

Premise 1: *If the alarm goes off, then the house is not secure.*

Premise 2: *The house is secure.*

Conclusion: *Therefore, the alarm did not go off.*

Take a look at this argument and see if you can identify which rule of inference it follows.

Modes Tollens

Problem

Premise 1: p or q .

Premise 2: *not* q .

Premise 3: *if* p , *then* r .

Conclusion: *Therefore*, r

Take a look at this argument and see if you can identify which rule of inference it follows.

Problem

Premise 1: p or q .

Premise 2: $\text{not } q$.

Premise 3: $\text{if } p, \text{ then } r$.

Conclusion: $\text{Therefore, } r$

Take a look at this argument and see if you can identify which rule of inference it follows.

r is true (Disjunctive Syllogism rule in first two premises)

Constructive Dilemma

Rule:-

Premise 1: p or q

Premise 2: if p , then r

Premise 3: if q , then s .

Conclusion : Therefore, r or s .

Example:-

Premise 1: *The weather is either sunny or rainy and*

Premise 2: *If it's sunny, we'll have a picnic.*

Premise 3: *If it's rainy, we'll watch a movie.*

Then:

Constructive Dilemma

Rule:-

Premise 1: p or q

Premise 2: if p , then r

Premise 3: if q , then s .

Conclusion : Therefore, r or s .

Example:-

Premise 1: *The weather is either sunny or rainy and*

Premise 2: *If it's sunny, we'll have a picnic.*

Premise 3: *If it's rainy, we'll watch a movie.*

Then: ***Either we will have a picnic or we will watch a movie.***

Problem

Premise 1: *If Alex is studying, then she is in the library.*

Premise 2: *If Alex is in the library, then she is quiet.*

Premise 3: Alex is studying.

Conclusion: ?

Problem

Premise 1: *If Alex is studying, then she is in the library.*

Premise 2: *If Alex is in the library, then she is quiet.*

Premise 3: Alex is studying.

Conclusion: ***If Alex is studying, then she is quiet.***

We used **Modus Ponens** to reach the final conclusion. By combining our new statement, If Alex is studying, then she is quiet , with the third premise, Alex is studying , we can logically conclude that ***Alex is quiet.***

We have proved successfully the argument is valid by using two rules of inference: the **Law of Syllogism** in the first step and **Modus Ponens** in the second.

Argument Form (Modus Ponens)

$$p$$
$$p \rightarrow q$$
$$\therefore q$$

Argument Form (Modus Ponens)

$$\begin{array}{c}
 p \\
 p \rightarrow q \\
 \hline
 \therefore q
 \end{array}
 \quad \Rightarrow \quad
 (p \wedge (p \rightarrow q)) \rightarrow q$$

Argument Form (Modus Tollens)

$$\begin{array}{c}
 p \rightarrow q \\
 \neg q \\
 \hline
 \therefore \neg p
 \end{array}
 \quad \Rightarrow$$

Argument Form (Modus Tollens)

$$\begin{array}{l}
 p \rightarrow q \\
 \neg q \\
 \hline
 \therefore \neg p
 \end{array}
 \quad \Rightarrow \quad
 ((p \rightarrow q) \wedge \neg q) \rightarrow \neg p$$

Proofs We have seen

AM $\geq$ GM - Algebraic Proof

To prove ,

$$(x + y) / 2 > \sqrt{xy}$$

when x and y are distinct real positive numbers, we can use **backward reasoning** .

We can construct a sequence of equivalent inequalities deduced from previous one.

$$(x + y)/2 > \sqrt{xy},$$

$$(x + y)^2 / 4 > xy,$$

$$(x + y)^2 > 4xy,$$

$$x^2 + 2xy + y^2 > 4xy,$$

$$x^2 - 2xy + y^2 > 0,$$

$$(x - y)^2 > 0.$$

Argument Vs Proposition

Statements in mathematics are called propositions.

In contrast,

An argument is sequence of statements with premises and a conclusion.

The last statement in an argument is called conclusion.

Whereas, all statements except last are called premises.

Argument - Example

Here is an example of argument.

- 1) If you are logged in on twitter, you can can access your twitter inbox.
- 2) You are logged in on twitter.

hence

- 3) You can access your twitter inbox.

Here (1) and (2) are **premises**. Whereas (3) is a **conclusion**.

Argument - Formal Notation

If we extract unique statements from previous examples, we get

- p = “You are logged in on twitter.”
- q = “You can access your twitter inbox.”

Then the argument in formal notation is written as

Argument Form

$$\begin{array}{c} p \rightarrow q \\ p \\ \hline \therefore q \end{array}$$

Statement Form

$$((p \rightarrow q) \wedge p) \rightarrow q$$

Create table

$$((p \rightarrow q) \wedge q) \rightarrow p$$

TABLE 1 Rules of Inference.

<i>Rule of Inference</i>	<i>Tautology</i>	<i>Name</i>	Example
$\begin{array}{l} p \\ p \rightarrow q \\ \hline \therefore q \end{array}$	$(p \wedge (p \rightarrow q)) \rightarrow q$	Modus ponens	You have an iphone. If you have an iphone, you are cool. You are cool.
$\begin{array}{l} \neg q \\ p \rightarrow q \\ \hline \therefore \neg p \end{array}$	$(\neg q \wedge (p \rightarrow q)) \rightarrow \neg p$	Modus tollens	If Raman is scientist, then Raman likes science. Raman does not like science. Raman is not scientist.
$\begin{array}{l} p \rightarrow q \\ q \rightarrow r \\ \hline \therefore p \rightarrow r \end{array}$	$((p \rightarrow q) \wedge (q \rightarrow r)) \rightarrow (p \rightarrow r)$	Hypothetical syllogism	If it snows, then it is cold. If it is cold, then you might get sick. If it snows, then you might get sick.
$\begin{array}{l} p \vee q \\ \neg p \\ \hline \therefore q \end{array}$	$((p \vee q) \wedge \neg p) \rightarrow q$	Disjunctive syllogism	Raman studies either Science or Maths everyday. Raman did not studied science today. Raman studied Maths today.

Logical Fallacies

Logical Fallacies

Logical **fallacies** are common mistakes in reasoning that lead to **invalid arguments**. Being aware of these helps avoid errors in mathematical proofs.

Example:-

- “If you don’t study, you fail the exam.”
- “You failed the exam”
- Therefore, “You must not have studied.”

Logical Fallacies

Logical **fallacies** are common mistakes in reasoning that lead to **invalid arguments**. Being aware of these helps avoid errors in mathematical proofs.

Example:-

- “If you don’t study, you fail the exam.”
- “You failed the exam”
- Therefore, “You must not have studied.”

This feels like valid argument, but it is not.

Here, the given student’s pen might have stopped working while writing exam.

So, always try to make valid arguments.

False Proof - The Infamous $1=2$ Proof

“Proof”: We use these steps, where a and b are two equal positive integers.

Step

1. $a = b$

2. $a^2 = ab$

3. $a^2 - b^2 = ab - b^2$

4. $(a - b)(a + b) = b(a - b)$

5. $a + b = b$

6. $2b = b$

7. $2 = 1$

Reason

Given

Multiply both sides of (1) by a

Subtract b^2 from both sides of (2)

Factor both sides of (3)

Divide both sides of (4) by $a - b$

Replace a by b in (5) because $a = b$
and simplify

Divide both sides of (6) by b

Logical Fallacies

- **Affirming the Consequent** : Proving the hypothesis if conclusion is true.

$$((p \rightarrow q) \wedge q) \rightarrow p$$

- *Example :-*

- If it rains, the ground becomes wet.
- The ground is wet.
- Incorrect conclusion: Therefore, it must have rained.
(perhaps someone watered the lawn!)

Logical Fallacies

- **Denying the Antecedent:** Proving the hypothesis if conclusion is true.

$$((p \rightarrow q) \wedge \neg p) \rightarrow \neg q$$

- *Example :-*
 - If you study, you will pass the exam
 - You didn't study.
 - Incorrect conclusion: Therefore, you won't pass the exam.
(perhaps you already know the material!)

Algebraic Proof

Prove that “The sum of two odd numbers is always even”.

Algebraic Proof

Prove that “The sum of two odd numbers is always even”.

Odd numbers are represented as $(2k + 1)$ for some integer k .

Let those two odd numbers be $(2a + 1)$ and $(2b + 1)$, for some integers a, b .

Sum:

$$\begin{aligned} \blacksquare & (2a+1) + (2b+1) = 2a + 2b + 2 \\ \blacksquare & = 2(a + b + 1) \\ \blacksquare & = 2m \end{aligned}$$

Since $(a + b + 1)$ is an integer, the sum is divisible by 2.

Hence, it is even.

Some Basic Terminology

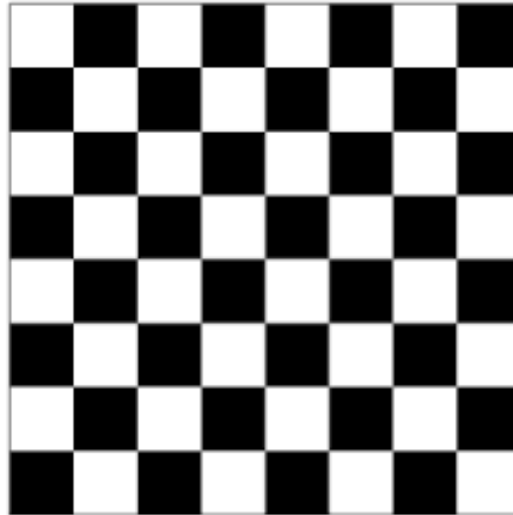
Theorem	A statement that can be true. Also known as facts or results.	Pythagorean Theorem: In a right triangle, $a^2+b^2=c^2$
Proof	A valid argument from known or established true statements. <i>Proof validates theorem.</i>	Pythagorean theorem proof using two different distribution of area
Axiom (Postulate)	A statement that is assumed to be true without any proof.	Euclid's Axiom : Through any two points, there is exactly one straight line.
Conjecture	A statement proposed to be true. Not yet proved. Becomes theorem after proved.	Goldbach's Conjecture: Every even number greater than 2 is the sum of two prime numbers

Some Basic Terminology

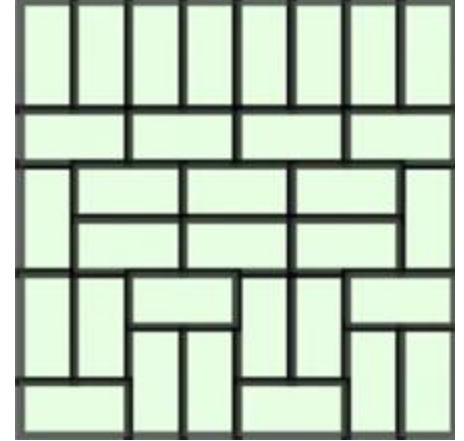
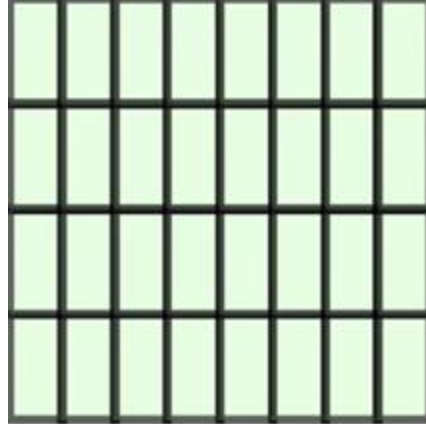
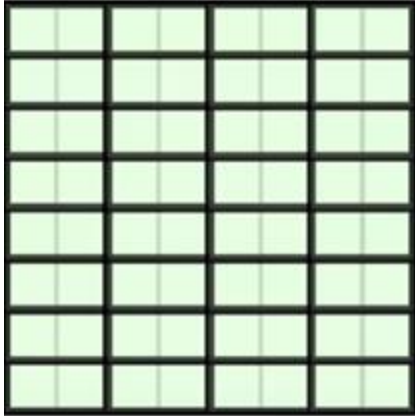
Lemma	Less important theorem used to prove other results. Proved separately.	Euclid's Lemma: If a prime p divides ab , then p divides a or b . Used in proving the Fundamental Theorem of Arithmetic .
Corollary	Another theorem directly proved from previously proved theorem.	Theorem: The sum of the angles in a triangle is 180° . Corollary: Each angle in an equilateral triangle is 60° .

Example:

Tile the chessboard with 1×2 tiles.

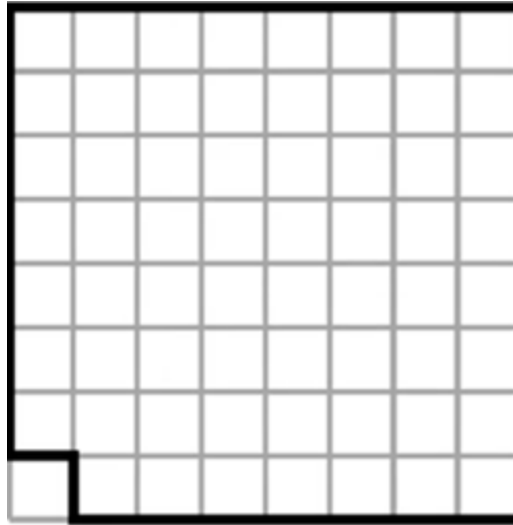


Proof by Example:



Example:

Tile the chessboard with 1×2 tiles.





**See You Guys in
Next Session :)**